



Research paper

Effective incentive based demand response with voltage support capability via reinforcement learning based multi-agent framework

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ABSTRACT

With the ongoing advancements in power grids, it is crucial to maintain a balance between energy generation and usage to ensure the stability of power systems. Any disparity between the supply and demand of energy can lead to increased expenses for utility companies and consumers, which might potentially result in a widespread failure of the grid. To tackle this problem, this study offers the implementation of the Voltage Capability Incentive-Based Demand Response (VC-IBDR) system, which utilizes a multi-agent framework. The main goal is to reduce power usage by providing financial incentives, while also ensuring that voltage standards are satisfied at each node. In this method, the aggregator agent coordinates demand response operations by interacting with household agents that are equipped with home energy management systems. Every household utilizes a disjunctively constrained knapsack problem optimization method to effectively organize the consumption of home appliances, considering customer preferences and dissatisfaction. During demand response events, the aggregator agent works together with household agents to optimize appliance schedules, ensuring that voltage levels at each node remain within acceptable levels. The simulation is performed to assess the efficacy of VC-IBDR on a feeder that provides service to 25 residential households. The findings indicate substantial decreases in power usage, with an average decline in demand of 4.20% at the feeder level and a 10.41% decrease in peak power consumption throughout the day. Furthermore, the voltage values at each node stay consistently steady within the range of 0.96 to 1.0 per unit throughout the day.

1. Introduction

In today's power grid, the persistent growth in energy consumption imposes significant strains on the power system. Any discrepancy between the electricity demanded and supplied can lead to substantial costs for both service providers and consumers, potentially causing severe disruptions to the entire power grid (Nojavan et al., 2017). As a result, service providers are compelled to secure ample energy resources to manage power fluctuations and enhance grid stability. Typically, conventional power sources are relied upon to address surpluses. However, this approach entails significant external factors and results in substantial carbon emissions (Allasseri et al., 2017). Demand Response (DR) initiatives present a promising approach to addressing the disparity between electricity demand and supply. Typically, DR schemes fall into two primary categories: price-based and incentive-based. Price-based DR encourages consumers (CUs) to adjust their energy consumption patterns in response to fluctuating power costs.

In contrast, incentive-based DR offers participating CUs constant or dynamic incentives to reduce their power usage (Shen et al., 2014). While price-based DR offers limited flexibility from the service provider's perspective, it also exposes CUs to potential challenges stemming from wholesale power price fluctuations. CUs wary of risk may hesitate to engage in such programs (Institute of Electrical and Electronics Engineers and IEEE Communications Society, 2016). Conversely, incentive-based DR is adaptable, providing greater flexibility for service providers to secure the necessary DR resources (Eissa, 2018). Although DR, particularly through incentive-based programs, has proven effective in managing flexible industrial needs, residential CUs have been slow to embrace this opportunity, despite their significant contribution to overall power consumption (Wang et al., 2020b). However, engaging residential customers in incentive-based DR (IBDR) operations poses challenges, as residential participants often contend with issues such as minimum active load requirements, response time constraints, and

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Nomenclature

x_a	Decision variable
ϕ_a	Dissatisfaction factor
π	Policy
ρ	Weighting factor
a	Appliance
A_n	Set of appliances of n household
AA	Aggregator agent
b^{HA}	Financial reward
C_a^{PS}	Dissatisfaction cost of PS appliance
C_a^{TS}	Dissatisfaction cost of TS appliance
D_T	Total demand
$F_{n,T}$	Forecasted power consumption of n household at time step T
H	Set of households
HA	Household agent
i_r	Incentive rate
I_{AA}	Incentives for AA
I_{HA}	Incentives for HA
$I_{T,n}$	Incentives of n household at time step T
$L_{p,a}$	Curtailment of power p of appliance a
n	Number of household
NS	Non Shiftable
P_a	Power demand of appliance a
P_T	Total demand at time step T
P_{NS}	Power demand of NS appliances
P_{PS}	Power demand of PS appliances
P_{TS}	Power demand of TS appliances
PS	Power Shiftable
R	Demand Reduction
Re^{AA}	Reward of aggregator agent
Re^{HA}	Reward of household agent
S_T^{AA}	State space for AA
$S_{n,T}$	State of the appliance n at time step T
T	Time step
T_i	Initial time step
TS	Time shiftable

privacy concerns, which are less pronounced in industrial settings (van Tilburg et al., 2023). Fig. 1 depicts a hybrid power distribution network that combines urban and rural areas, incorporating both centralized and distributed renewable energy sources. Households in both urban and rural areas that have wind turbines and solar cells contribute to the distributed generation of electricity, highlighting the network's dependence on renewable energy sources. Furthermore, it shows an example of an active distribution network with local intermittent generations beside consumers and most of the power consumption is residential users. Initially, it may be more practical to incorporate residential users into the IBDR program as individual entities, with their loads aggregated for streamlined management.

Besides, aggregators play a pivotal role in the energy market, acting as intermediaries between the Distribution System Operator (DSO) and the customer. They facilitate residential participation in IBDR programs (Zhou and Yang, 2018). Moreover, the sudden emergence of demands for load reductions within IBDR programs necessitates swift responses, a task that residential participants may struggle to manage effectively. One effective solution to this challenge is to implement automated responses at the consumer's end using a Home Energy Management System (HEMS). Reinforcement learning (RL) enhances decision-making processes by actively engaging with the environment,

thereby reducing the reliance on prior information, resolving uncertainty autonomously, and being relatively straightforward to implement in real-world scenarios. Numerous studies advocate for the use of RL in DR within the residential sector. Notably, a significant portion of these studies focus on managing energy consumption within individual households. Mathew et al. (2020) introduces a method for implementing residential DR by employing deep RL to optimize aggregated energy consumption based on time of use (TOU). However, the case studies in the paper are confined to just five consumers, potentially limiting the generalizability of the findings to larger populations. Additionally, the absence of appliance scheduling and the oversight of consumer dissatisfaction in the author's analysis are noteworthy limitations. The author of Lu and Hong (2019) introduces a real-time RL algorithm tailored for IBDR programs, aimed at aiding service providers in acquiring energy flexibility from residential customers. This algorithm employs a single-agent RL approach to compute nearly optimal incentives for diverse consumers, striking a balance between participant benefits and discomfort. However, the author overlooks the integration of the HEMS system for the customer. In Chung et al. (2021), the author presents a system that does not rely on a specific model to schedule appliance consumption patterns across multiple homes. Instead, households are depicted as non-cooperative stochastic games, with RL employed to determine the Nash equilibrium while safeguarding consumer privacy. However, this approach overlooks considerations of customer satisfaction when scheduling appliances. The authors of Lu et al. (2019) introduce an hour-ahead energy consumption scheduling challenge, accurately represented as a finite Markov decision process with discrete time intervals. This study exclusively focuses on implementing HEMS for a household to optimize appliance usage and minimize energy consumption during periods of high electricity prices. However, it is constrained to a single household HEMS-based DR framework. In Alfaverh et al. (2020), the paper presents a household HEMS employing Q-learning to effectively address the complexities arising from fluctuating electricity costs and varying power usage patterns, while ensuring alignment with user preferences. By incorporating user feedback into decision-making logic through fuzzy logic reward functions, the system adjusts for human preferences. Nonetheless, the scope is limited to single-household HEMS for TOU-based appliance scheduling. The authors of Liu et al. (2019) introduce a comprehensive pricing-based DR system for smart homes, accommodating various appliance types and prioritizing consumer satisfaction. The proposed architecture facilitates the flexible integration of household appliances with diverse features, as well as energy storage units and distributed energy resources. However, the study employs a Mixed Integer Linear Programming (MILP) based HEMS framework for a single household, which encounters challenges in handling extensive state and action spaces and may require precise system dynamics modeling. The RL technique offers advantages in real-world scenarios. Khan et al. (2022) demonstrates a multi-agent RL system tailored for TOU, integrating both DR and voltage control. Employing a Long Short-Term Memory (LSTM) network for day-ahead load forecasting mitigates future risks, while Q-learning enables effective demand response. However, the simulation overlooks customer dissatisfaction regarding power curtailment. The authors of Si et al. (2021) present an algorithm for HEMS capable of predicting and responding to energy demand changes one hour in advance. Utilizing an artificial neural network for price prediction addresses pricing unpredictability. The decentralized approach, driven by multi-agent RL, optimizes appliance selections. However, the algorithm is not extended to multiple houses. The authors of Zhang et al. (2021) outline a framework for DR control fostering mutually beneficial connections among utility companies, buildings, and cloud service providers. By formulating a Markov Decision Process, the study addresses air conditioning system DR in small and medium commercial buildings, integrating a nonlinear reward structure to balance comfort and energy consumption. Nonetheless, it focuses solely on air conditioning DR for buildings. Xu et al. (2021) presents a comprehensive

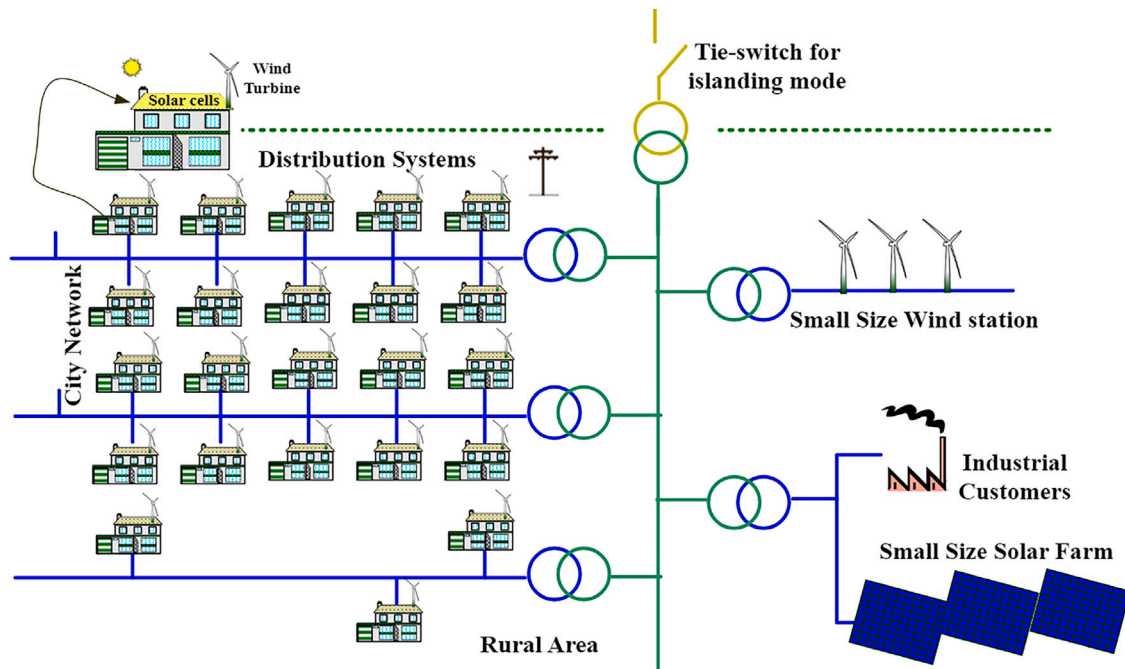


Fig. 1. An active distribution network with household agents.

market structure enabling residential consumers to actively engage in the day-ahead market through proactive demand response activities. Utilizing agent-based modeling and simulation, the study assesses interactions among generating firms, retailers, residential consumers, and the independent system operator. Xu et al. (2020) presents HEMS system that can efficiently schedule electric appliances based on fluctuating electricity prices by utilizing Q-learning to understand customer preferences and make decisions for rescheduling of the appliances. The author only focused on HEMS system within a household. The paper Li et al. (2020b) introduces a real-time DR method that utilizes deep RL and policy search algorithms to optimize the arrangement of home appliances. This technique considers uncertainty of the resident's behavior, power price, and the outside weather. This study is limited to a single household. Chen et al. (2021) and Li et al. (2020a) present a multi-armed bandit method and its variants to utilize for choosing the right consumers for DR participation during the preparation time, when customer actions are uncertain. However, in the context of real-time load control in IBDR, these studies overlook to consider or oversimplify the uncertainty associated with consumer behaviors. Additionally, the impact of numerous environmental elements is typically neglected. Authors of Zhang et al. (2022) present a residential DR strategic bidding method for load aggregators to enhance system reliability and optimize their financial benefits using deep reinforcement learning. However, Author do not consider the HEMS system for appliance scheduling for DR action. Sun et al. (2024) presents a closed-loop reverse Stackelberg game-based pricing mechanism for more effective regulation of DR program. In the hierarchical game structure, the lower-level consists of the non-cooperative competition among different demand-side resources in participating DR. The upper level is determined by the pricing policy set by the load aggregators. The author applied DR on an aggregated consumption and does not consider the different appliances. Shi et al. (2020) introduces the connections between household appliance activities, such as electric water heaters and air conditioners, load profiles, and participation in the IBDR programs. This study only focuses on rewards for reducing peak electricity usage that may effectively reduce the overall demand on the electrical grid. The authors of Yu et al. (2020) introduce a sophisticated deep deterministic policy gradient algorithm for optimizing DR strategies in the control of different household appliances. Mocanu et al. (2019) introduced an approach for efficiently managing energy consumption by

scheduling the time-scaling and time shifting load. This strategy utilizes advanced algorithms such as deep Q-networks and deterministic policy gradient to achieve optimal energy optimization of a building. Wang et al. (2020a) and Vázquez-Canteli and Nagy (2019) introduce a control agent characterized by its data-driven, model-free, and closed-loop nature. This agent is trained using deep RL for voltage control, yet it does not integrate DR.

In the existing literature, DR applications predominantly focus on single households or groups of households, overlooking studies for a residential sector like a feeder level. However, when implementing DR across approximately 25 houses connected by a radial feeder, the risk of voltage violations arises, requiring an active multi-agent system to mitigate such occurrences. Table 1 offers a comparative overview of research endeavors in both the demand response domain and voltage control within distribution networks. Notably, there is a gap in research that applies the IBDR program to a residential distribution feeder, specifically focusing on the use of RL for incentive-based residential DR, while concurrently applying voltage control to optimize the standard voltage throughout the feeder. Most studies in this domain predominantly focus on optimizing energy or implementing demand response within single households without considering voltage violations in a distribution network.

To fill this research gap, this work presents a novel incentive based demand response with voltage level support capability. The key contributions of this study are as follows:

- The VC-IBDR environment involves real-time decision-making for both aggregators and individual household participants, maximizing benefits such as financial incentives for households and reduced power consumption demand for the supplier.
- The HEMS system is developed for each household connected to the feeder to take DR actions, taking into account each consumer's dissatisfaction level with scheduling the appliances and weather considerations.
- Propose the decentralized multi-agent RL (MARL) based model-free approach for the decision-making of home appliances and to make communication only with the aggregator agent without sharing the information with the rest of the participants agents.

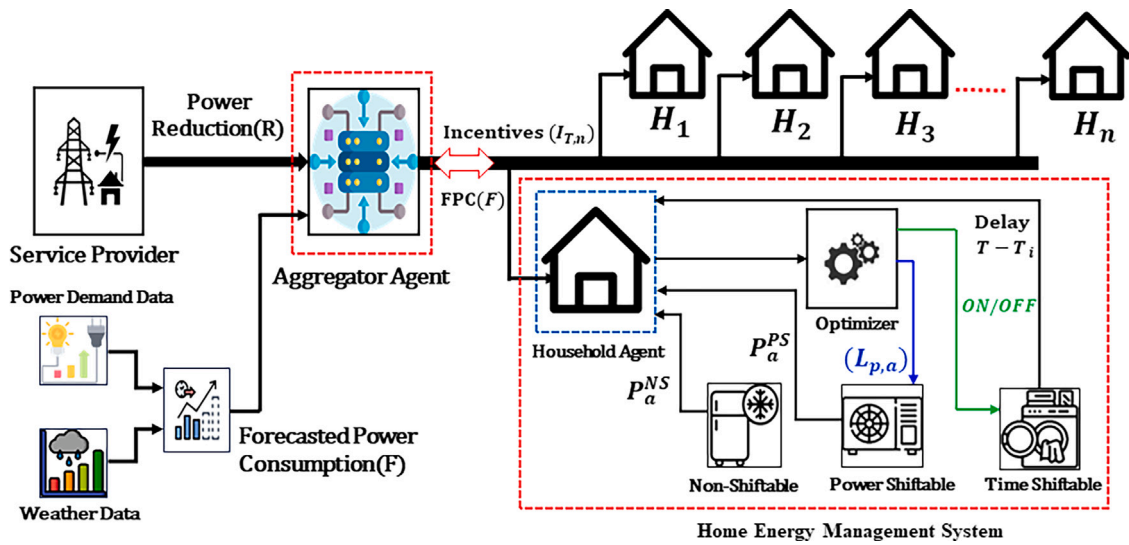


Fig. 2. Proposed Architecture for Incentive Based Demand Response.

Table 1
Overview of Related Work and the Proposed Method.

Ref.	Customer preference	Multiple houses	Weather consideration	Voltage control	Incentive/ Time of use based
(Mathew et al., 2020)	X	✓	X	X	TOU
(Lu and Hong, 2019)	✓	✓	X	X	IB
(Chung et al., 2021)	X	✓	X	X	TOU
(Lu et al., 2019)	✓	X	X	X	TOU
(Alfaverh et al., 2020)	✓	X	✓	X	TOU
(Liu et al., 2019)	X	✓	X	✓	TOU
(Khan et al., 2022)	✓	X	X	X	TOU
(Si et al., 2021)	✓	X	✓	X	TOU
(Zhang et al., 2021)	✓	✓	✓	X	TOU
(Xu et al., 2020)	✓	X	X	X	TOU
(Li et al., 2020b)	✓	X	✓	X	TOU
(Chen et al., 2021)	X	✓	X	X	IB
(Li et al., 2020a)	X	✓	X	X	IB
(Zhang et al., 2022)	X	✓	X	X	IB
(Sun et al., 2024)	X	✓	X	X	IB
Proposed	✓	✓	✓	✓	IB

- Propose the method to mitigate the rebound effect of peak demand occurring in the time slots where power consumption is high.
- Propose the voltage control in the MARL architecture to maintain the voltage level within the acceptable range for each household connected to a feeder.

2. Problem formulation

This study specifically concentrates on the IBDR algorithm that gives financial incentives to the consumers as a reward for reduction of power consumption along with voltage control mechanism which ensures the acceptable range of voltages at each node. In this study, an aggregator agent (AA) gets input of power reduction and forecasted/reference power consumption of a particular day and it sends incentives and reference consumption pattern in a way that a consumer gets incentives for reducing its consumption below the reference power demand line. A household agent (HA) performs a home energy management for demand response according to the user's demand and their dissatisfaction with power shifting or curtailing of appliances. AA receives the voltages of all the HAs at each time instant to check the voltage level compliance with the standard. If the voltages are out of acceptable range at any node then it sends command to HAs to curtail the further load according to the voltage deviation. Fig. 2 shows the proposed architecture for a smart grid structure in which the service provider, aggregator agent, and household agents

collaborate to effectively control and enhance energy usage. Incentives are employed to motivate households to take part in DR programs, while the optimizer integrated into the HEMS ensures optimal energy consumption by categorizing loads and making well-informed choices on the shifting or delaying of power usage. This strategy improves the reliability as well as efficiency of the power distribution system, particularly during times of high demand and Fig. 3 shows the proposed voltage control mechanism, which is an integrated methodology for monitoring and controlling voltage levels in a smart grid. AA identifies the most sensitive voltage violation node for further DR actions through the HEMS system. If the voltages remain within the range throughout the feeder, then it quits the further DR actions; otherwise, it will continue to request the next nearest node to lower the power consumption. The mathematical formulation for the voltage control, AA and the HEMS for each HA as well as the objective function are thoroughly explained in the subsequent sections.

2.1. Home energy management system

The HA performs all the actions within the house to participate in the IBDR program. The number of households n participating in the DR program is represented by the set of all households $H = \{1, 2, 3, 4, \dots, n\}$. Each HA controls the set of appliances: Power-shiftable (PS), Time-shiftable (TS) and Non-shiftable (NS), $A_n = \{PS, TS, NS\}$ in such a way that it maximizes the financial reward and minimizes the demand by considering the user's dissatisfaction due to the shifting or

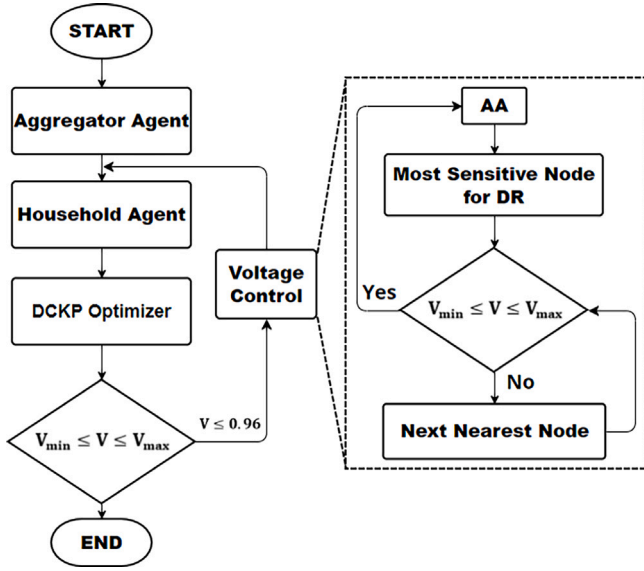


Fig. 3. Proposed Voltage Control Mechanism.

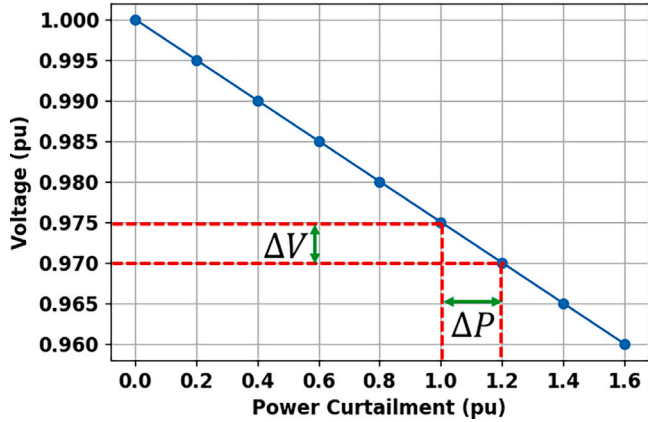


Fig. 4. VP droop characteristic curve.

Table 2
Appliances and their Parameters.

Appliance	Type	Dissatisfaction	Demand (kW)
Washing machine (WM)	TS	0.10	1.0
Dryer	TS	0.20	2
Dishwasher (DW)	TS	0.06	2
EV	TS	0.05	4
AC1	PS	0.30	0–4
AC2	PS	0.40	0–4
Non-Shiftable	NS	1.00	0–5

curtailing the appliances. The environment model for each household agent is determined by its state, the available actions, the rewards associated with those actions, and the optimizer that determines the shifting, delaying, or curtailing of the appliances. These appliances are divided into three types: PS, NS and TS respectively. Each type of appliances is the subset of appliances set A_n and it makes the state for the RL model $A_n = PS_n \cup NS_n \cup TS_n$

2.1.1. Non-shiftable appliances

Non-shiftable appliances are distinguished by necessary demand requirements that must be fulfilled throughout the power distribution process. When they start operation, it is required to operate continuously and cannot be subject to schedule. The power consumption of

such appliances $a \in A_n$ is always equal to power demand. These types of appliances participate as a single state for RL environment.

2.1.2. Time-shiftable appliances

Time-shiftable loads can adjust their power consumption to another time step where demand is low. This not only helps to prevent excessive consumption of electricity during times of high demand, but also results in a reduction in the energy bill. Time-shiftable appliances possess two distinct operational states, namely “on” and “off” and these devices are further divided into two categories: interruptible (devices can stop their operation and continue later like EVs) and non-interruptible (devices which complete their operation without interruption like dishwasher). Their state is defined by the time difference between the current time step (T) and the initial time steps (T_i) and the delay time will be $T - T_i$.

2.1.3. Power-shiftable appliances

The power shiftable loads have the capability to be operated with a variable power consumption that ranges between the minimum and maximum power demands and they do not allow any delay in usage. Their state is defined as power consumption demand.

The current state ($S_{n,T}$) of an HA at time step (T) encompasses the information from all appliances $a \in A_n$. Furthermore, the HA continuously tracks the incentives at time step (T) for (n) household is $I_{T,n}$ that are determined by the AA and its own reference power consumption ($F_{T,n}$). Therefore, the observation for HA (n) in time step (T) is given in equation.

$$O_{T,n}^{\text{HA}} = \{S_{n,T}, F_{T,n}, I_{T,n}\} \quad (1)$$

Action space for this problem is defined such a way that HA compares the reference demand (F_T) with the consumer's total demand (P_T) at time step (T). If power consumption (P_T) is greater than reference power demand (F_T), then it will reduce the power consumption according to user's preferences using HEMS and HA is rewarded by AA with incentives (I_{HA}) and dissatisfaction cost (C) caused by the power reduction. It always tries to maximize power reduction when it exceeds the reference demand and minimize dissatisfaction. The financial reward is calculated by following equation.

$$b_{HA} = \max \sum_{T=1}^{T=96} [F_{T,n} - P_{T,n}] \quad (2)$$

Shifting or curtailing power demand causes dissatisfaction. In the case of a power-shiftable appliance, the dissatisfaction cost (C_a^{PS}) is a function of power curtailment ($L_{p,a}$) and the dissatisfaction factor (ϕ_a). Power is curtailed by 0.2 kW level in between minimum and maximum power demand of the appliance set by the user and it is given in Eq. (3). In the case of time shiftable, dissatisfaction cost (C_a^{TS}) is function of time delay, shown in the following equation.

$$L_{p,a} = [P_{a,\min} : 0.2 : P_{a,\max}] \quad (3)$$

$$C_a^{PS} = \phi_a \cdot (L_{p,a}) \quad (4)$$

$$C_a^{TS} = \phi_a \cdot (T_n - T_{i,n}) \quad (5)$$

The total reward function for the HA comes by combining the financial reward and dissatisfaction cost which is presented as:

$$\text{Re}^{HA} = \max \sum_{T=1}^{T=96} [F_{T,n} - P_{T,n}] - [C_a^{PS} + C_a^{TS}] \quad (6)$$

HA agents receive financial incentives for reducing their power consumption, according to following equation:

$$I_{HA} = \frac{P_{\text{total (without DR)}} - P_{\text{total (with DR)}}}{P_{\text{total (without DR)}}} \cdot I_{AA} \quad (7)$$

where I_{AA} is the total incentives given by the AA and I_{HA} is the incentives received by each HA

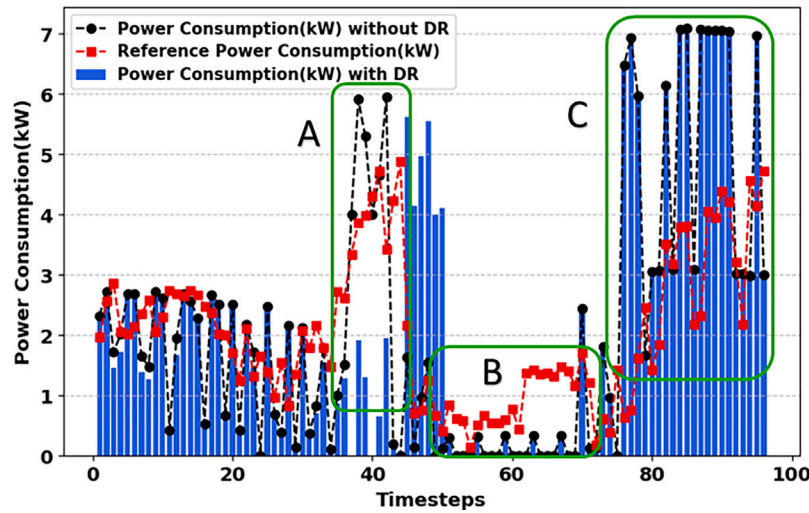


Fig. 5. Rebound effect with DR.

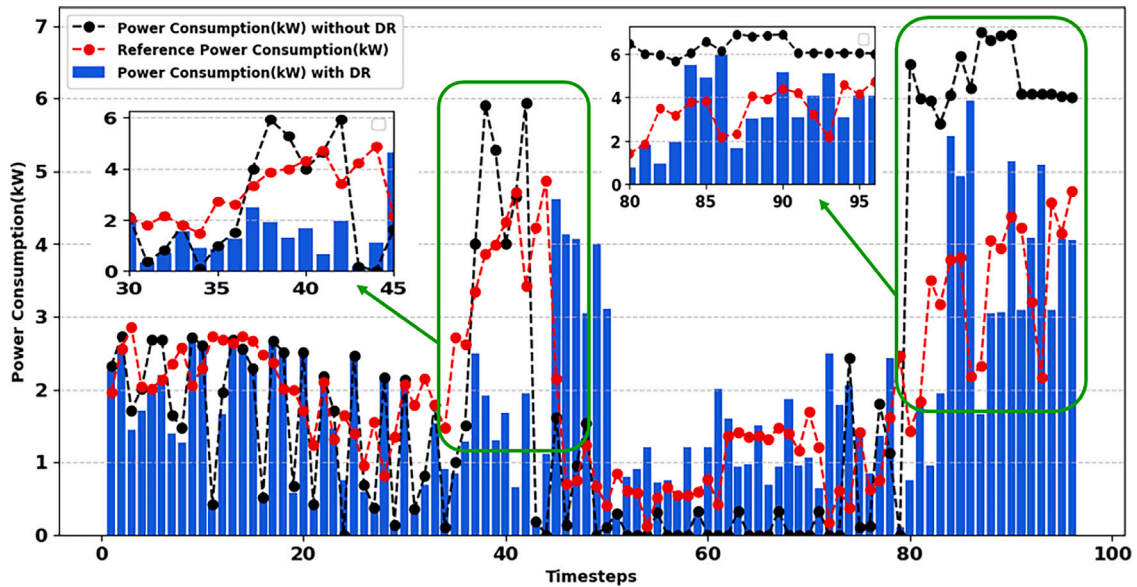


Fig. 6. Power Consumption Pattern of a Household with and without DR.

2.2. Disjunctively constrained knapsack problem based optimizer (DCKP)

Optimizer within the HA performs the function of optimal curtailing or shifting of appliances based on reference demand. The optimizer uses DCKP in such a way to schedule the devices based on customer's dissatisfaction and power consumption with the objective to minimize the power demand less than the demand limit and it is given in the following equations:

$$\min \left\{ \sum_{a \in PS} C_a^{PS} + \sum_{a \in TS} x_a \cdot C_a^{TS} \right\} \quad (8)$$

$$\text{subject to: } \sum_{a \in PS} P_a^{PS} + \sum_{a \in TS} x_a \cdot P_a^{TS} \leq F \quad (9)$$

where $x_a \in \{0, 1\}$ is used for decision variable for ON and off of TS appliances. After solving the DCKP, power demand for PS and TS is presented as:

$$P_T = \sum_{a \in TS} P_{TS} + \sum_{a \in PS} P_{PS} \quad (10)$$

2.3. Rebound effect consideration

Rebound effect is the shifting of appliances in such time steps where power consumption is high causes the second peak in that time slot which is undesirable effect that can leads to instability of the grid. To mitigate the rebound effect, appliances are scheduled in such a way that minimizes the difference between total demand and the reference demand given by AA while considering the power consumption of each appliance, presented in the following equations:

$$\min \left\{ \sum_{a \in PS} P_a^{PS} + \sum_{a \in TS} x_a \cdot P_a^{TS} \right\} \quad (11)$$

$$\text{subject to: } P_a^{NS} + \sum_{a \in PS} P_a^{PS} + \sum_{a \in TS} x_a \cdot P_a^{TS} \leq F_T \quad (12)$$

2.4. Aggregator agent (AA)

The state space of an aggregator agent comprises of two states: total demand (D_T) and reduction demand (R). State space for AA is ($S_T^{AA} =$

$\{D_T, R\}$. AA sends demand reduction and the associated incentive rate (i_r) to HA to perform the action in each time step and receives reward as given in equation:

$$Re_T^{AA} = -[\rho \cdot D_T^+ + (1 - \rho) \cdot \sum_{n \in H} b_{HA}] \quad (13)$$

AA agent considers a penalty for exceeding target reduction. The first term of Eq. (13) shows the penalty and the second term is for total incentive paid to HA agents. There is a weighting factor ρ which defines the trade of between two terms. There is no reward for AA if aggregated demand is lower than target reduction as the aim of DR program is to reduce power consumption to meet the capacity constraints of service provider. Total incentives paid by AA are calculated by using the following equation:

$$I_{total}^{AA} = \frac{\text{Demand overflow (kW)}}{\text{Demand Range (kW)}} \times \max(i_r) \quad (14)$$

where demand range is the difference between maximum total demand and capacity threshold which is defined by the service provider and demand overflow is the difference of total demand and capacity threshold.

2.5. Voltage control mechanism

HA agents perform DR actions by using the HEMS system, and then AA agent monitors the voltage levels at each household node across the distribution network at each time step. If the voltage level is found to be less than ≤ 0.96 then it will request the respective HA agent to curtail the further load according to the voltage droop characteristic curve given in Fig. 4 to raise the voltage in such a way, first it will identify the most sensitive node and calculate the ΔV corresponding to load curtailment ΔP . The node sensitivity is presented by the following equations:

$$[S_{IP}] = \frac{\partial V_i}{\partial I_{P_j}} = -[R] \quad (15)$$

$$[S_{IQ}] = \frac{\partial V_i}{\partial I_{Q_j}} = -[X] \quad (16)$$

The sensitivity matrices related to the real and reactive part of current are denoted as S_{IP} and S_{IQ} respectively. On the other hand, R and X represent the resistance and reactance components in the impedance matrix [Z] (Arshad et al., 2018).

3. Multi-agent reinforcement learning methodology

In this work, multi agent reinforcement learning (MARL) is applied to train HA and AA simultaneously as they have opposite reward function indirectly. Therefore, the agents have the challenge of dealing with a dynamic situation where the most effective policy evolves as opposing individuals modify their own strategies. The MARL algorithm efficiently balances the exploration and exploitation trade-off that employs the action-selection strategy of ϵ -greedy with decay. This means that, given a certain probability ϵ the agent chooses a random action. The value of ϵ lowers over time with a decay rate δ . MARL utilizes an approach that involves exploration during the initial phases of training and then improving the policy throughout subsequent stages. Deep Q-Networks (DQNs) are used by MARL to account for large and continuous state spaces that are impractical for Q-learning (Crites and Barto, 1995). To enhance the stability of training using DQNs, two specific aspects are employed: (i) To prevent non-stationary targets, a distinct target network σ^T is employed to establish the target values, while the original network is utilized for Q-value prediction (ii) Utilize replay buffer B mechanisms to prevent the agent from disregarding past experiences. Every experience is stored in a replay buffer B . The network is trained on the batches that are randomly selected for experience from B as opposed to just using the most recent experience. Algorithm 1 presents the training technique for MARL for DR. At the

beginning of the method, a policy network and target network are set up for each individual agent. Subsequently, all agents undergo training to optimize the networks over a specific number of episodes. Each episode represents a single day and consists of a sequence of T time steps. At each time step T , the AA chooses an action. After that, each individual HA (n) chooses an action and promptly receives their corresponding reward. Once all HAs have determined their reaction to the AA, the reward for the AA may be computed. The training process requires a substantial time investment to acquire policies for each agent. After being trained, the RL agent may be effectively utilized in real-time by employing policy π_s given in Eq. (17).

$$\pi_s = \arg \max Q(s, a | \sigma) \quad (17)$$

4. Results and discussion

This section presents the simulation results for VC-IBDR using MARL to evaluate the performance of AA as well as HAs, integrating with the voltage control. In this simulation, 1-year, 25 households power consumption data for each household's appliance from the Pecan Street datasets, is used to evaluate the proposed algorithm (Anon, 2023). The VC-IBDR algorithm is trained for 3000 episodes in a decentralized manner for a day, which is divided into $T = 96$ time steps with a 15-min resolution for 1440 min. Model is trained on the sample data between Jan 1, 2018 to September 30, 2018 to consider both summer and winter months. AA and HA agents have individual DQN environments to take actions according to user's dissatisfaction coefficients (ϕ_a) which is set by the user. The appliance type, demand and dissatisfaction are given in Table 2 and MARL optimization parameters are set which includes learning rate $\eta = 0.01$, Discount factor $\gamma = 0.9$, the decay rate $\delta = 0.99$ are set in ϵ -greedy to converge the learning rate over the time. Weighting factor ρ is set at 0.6 and the demand reduction is set by the service provider according to the maximum demand and available capacity such a way that the maximum demand should be less than the capacity of the grid.

Algorithm 1 MARL based DR Training.

Input: Initialize DQN for σ^{AA} , $\sigma^{(T,AA)}$, B^{AA}
Input: Initialize DQN for σ^{HA} , $\sigma^{(T,HA)}$, B^{HA} for each $HA(n)$
for each episode **do**
 Initialize target reduction demand R
 Initialize rewards for AA and HA
 for each time step T **do**
 Forecast demand and set it as reference demand (F)
 $o_T^{AA} \leftarrow (D_T, R)$
 Add $(o_{T-1}^{AA}, I_{T-1}, Re_o^{AA}, Re_o^{HA})$
 Train σ^{AA} given Reference Power Consumption(F)
 Select I_r using ϵ -greedy
 for all $HAs, n \in H$ **do**
 Observe the appliances state ($S_{n,T}$) and set $F_{T,n}$
 Add $(o_{T-1,n}^{HA}, a_{T-1,n}, Re_{t-1,n}^{HA}, O_o^{HA})$ to B
 Train σ_n^{HA} given B
 Select $a_{T,n}$ using ϵ -greedy
 Obtain $P_{PS,n}$ and $P_{TS,n}$ by solving optimizer
 Update $s_{T+1,n}$ according to $P_{PS,n}$ and $P_{TS,n}$
 Calculate reward for HA ($Re_{T,n}^{HA}$)
 end for
 Calculate reward for AA ($Re_{T,n}^{AA}$)
 end for
end for

4.1. Household power consumption analysis

AA interacts with the HAs as the aggregated demand exceeds the capacity limit of the grid, and it sends requests to the HAs to lower the power consumption. Accepting the incentive rate, HA performs the

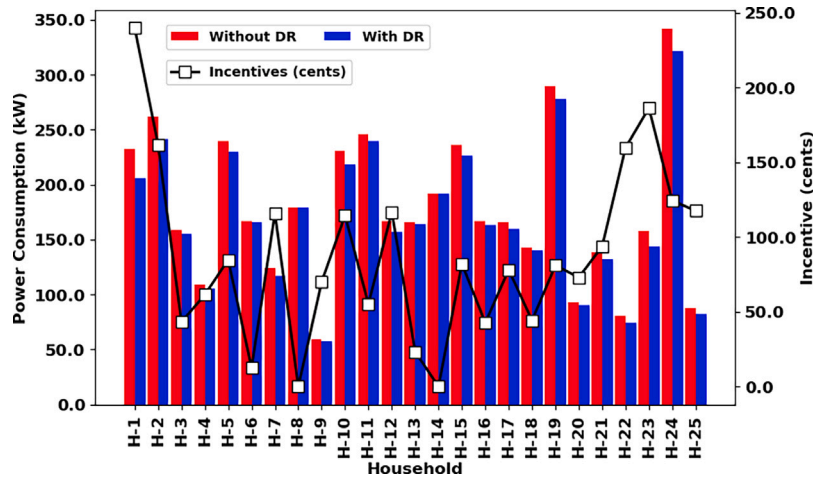


Fig. 7. Aggregated performance of an individual household for single day power consumption.

DR action in the HEMS system. Fig. 5 shows the performance of HEMS for a household. The highlighted area (A) shows the DR reaction as the actual power consumption surpasses the reference demand, and the optimizer schedules the appliances to lower the power demand at that time step. However, the optimizer fails to schedule appliances optimally because it shifts the power demand to the areas with high actual power consumption, leading to a rebound effect. This is evident in highlighted area (C), where (B) displays low actual power consumption in that time step because almost all the appliances were already off leads to lower power consumption. The optimizer effectively mitigates this rebound effect by scheduling the appliances during low-demand time slots. Fig. 6 shows the optimal DR actions performed by the HEMS and it reduces power in time steps between 30 and 45 to meet the DR action condition. The optimizer reduces power consumption and optimally schedules appliances during low demand periods, taking into account the rebound effect, as demonstrated in time steps 80–95. Fig. 7 displays the combined performance of each household's load consumption, both with and without DR, over a single day in June. H-1 reduces the power consumption by 11.55% and gets incentives accordingly, whereas H-2 participated in the DR to lower the power consumption by 7.79%. It is not necessary for every household to participate in the DR program as H-8 and H-14 do not participate and get zero incentives.

4.2. Aggregator agent performance analysis

VC-IBDR reduces the load and gives instructions to all HAs to participate in the DR program as the demand exceeds the capacity limit defined by the service provider. Fig. 8 depicts the performance of the AA in the VC-IBDR program for a day. Highlighted area (A) depicts the behavior of DR and the incentives enabled as the aggregated power reduction is done, while highlighted area (B) shows the peak reduction and manages power demand below the capacity limit in peak time. Highlighted area (C) illustrates the implementation of DR action to mitigate the rebound effect, which involves reducing the power demand and striving to stay close to the grid's capacity limit. The AA reduces the peak load by 7.82%, the mean load is reduced by 4.139%, and the peak-to-average ratio is improved by 2.18% with DR by giving incentives of 2076.37 cents to HAs, and the mean incentive received by each HA is 83.83 cents.

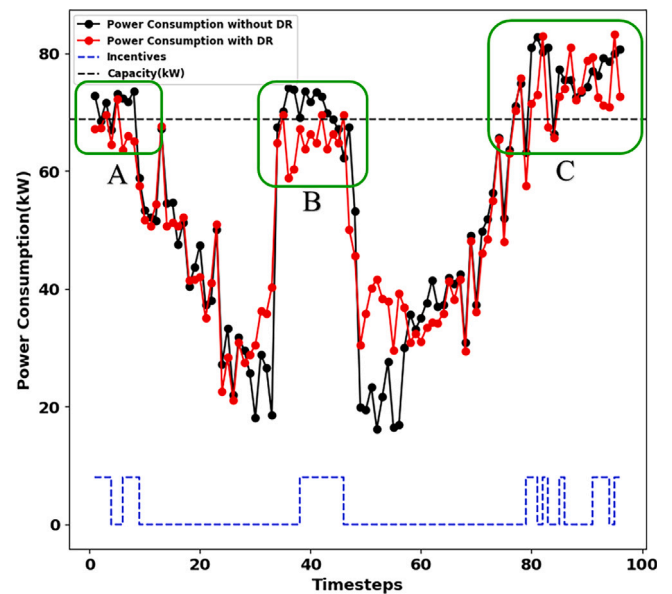


Fig. 8. Aggregated Power Consumption.

4.3. Performance of voltage control

Fig. 9 illustrates the impact of the multi-agent system on the voltage levels at node 24, regulating the voltage at each time step to ensure acceptable levels. The absence of the proposed mechanism in this study results in significantly low voltage levels. The DR algorithm, when combined with the voltage control, effectively regulated the voltage levels at each node for every time step, ensuring they remained within the specified limits, specifically $V \geq 0.96$ PU. Consequently, maintaining voltage levels within an acceptable range across the distribution network enhanced the effectiveness of the proposed system. Fig. 10 shows the average voltage deviation at each node with DR as well as with the voltage control mechanism. From nodes 1–12, average voltages are achieved due to almost consistent power demand. However, at the 12th node, the average voltage decreases suddenly because of the sudden increase in load at the feeder.

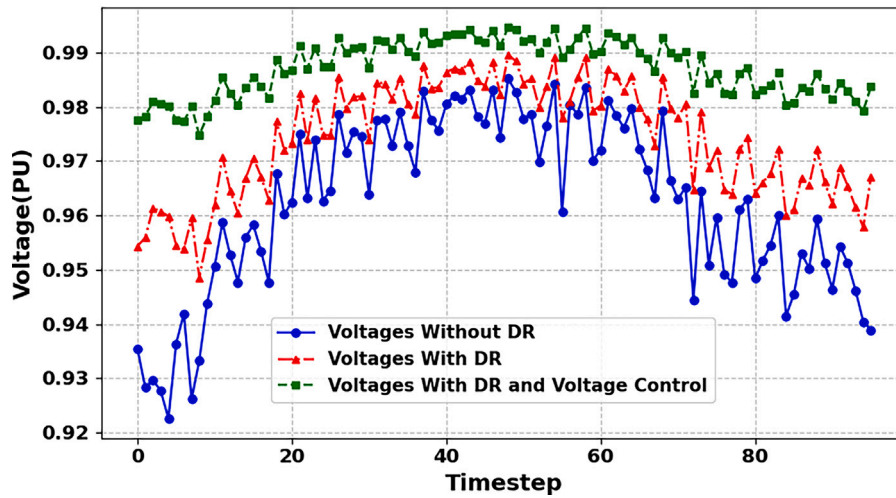


Fig. 9. Voltage Profile at Node-24.

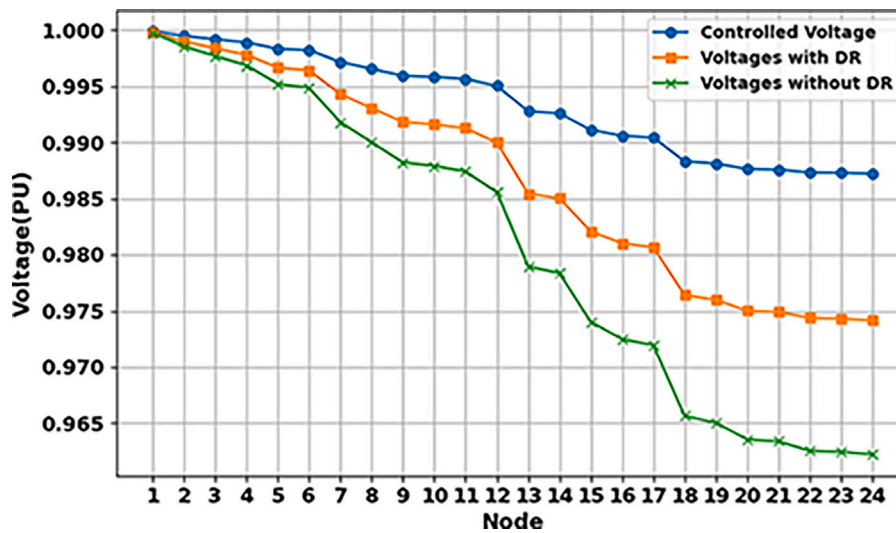


Fig. 10. Average Voltage at each node: without DR, with DR and with Voltage Control.

4.4. Analysis of MARL-DR performance

Table 3 shows the 1-year performance of the DR program for each month's per-day average peak load, power demand, and incentives paid. Owing to the fact that June is highly involved in the DR program because of the high demand for power consumption, per-day average peak load and demand reduction are observed by 7.06% and 4.71%, respectively, by paying 1973.02 € incentives. Furthermore, December is observed to be a less participated month in the DR program as average incentives given by AA are 295.64 € because DR program participation depends on the capacity limit, demand reduction, and incentive rate of the day, which varies day by day according to the weather condition and power demand.

5. Conclusion

Through the utilization of multi-agent reinforcement learning, this study offers the use of voltage control and incentive-based demand response. The approach is validated by the electricity demand of twenty-five households, which takes into account a dissatisfaction of those households with the weather conditions. The performance of the MARL-based algorithm is effectively demonstrated in the results, which indicate a reduction in average demand of 4.71% at the feeder and a reduction in average peak power of 7.06% achieved in a single day during the month of June. Additionally, the voltage is maintained at each node in an acceptable range of 0.96–1 PU for each time step. We evaluate a home energy management system of each household that incorporates DCKP optimization in order to schedule appliances

Table 3
One Year analysis of average Power Demand with and without DR and average incentives (¢) paid.

Month	Without DR		With DR		Incentive paid by AA	Incentive paid to each HA
	Peak load (kW)	Power demand (kW)	Peak load (kW)	Power demand (kW)		
Jan	26.13	12.97	24.99	12.55	504.47	20.17
Feb	28.06	13.65	27.06	13.25	323.40	12.93
Mar	34.95	15.04	33.36	14.68	341.83	13.67
Apr	39.43	15.16	35.57	14.54	621.84	24.87
May	74.58	33.09	67.79	31.69	1413.79	56.55
Jun	84.47	41.87	75.67	40.11	1462.43	58.49
Jul	86.72	46.69	80.52	44.49	1973.02	78.92
Aug	87.31	47.66	79.81	46.11	1621.22	64.84
Sep	69.62	33.40	63.71	32.34	786.18	31.44
Oct	46.61	20.49	43.41	19.98	355.66	14.22
Nov	36.27	18.69	34.25	17.86	330.22	13.25
Dec	30.56	15.91	28.39	15.11	295.64	11.28

according to reference demand. In addition to this, the voltage control system guarantees that the voltages at each node are within the permitted range. For the purpose to keep the grid stable, we use this innovative method to reduce the amount of energy demand and to adjust the voltages accordingly.

CRedit authorship contribution statement

Fayaz Ahmed: Writing – original draft, Validation, Software, Methodology, Formal analysis, Conceptualization. **Ammar Arshad:** Writing – review & editing, Visualization, Investigation, Formal analysis. **Attique Ur Rehman:** Writing – review & editing, Software, Resources, Data curation. **Mohammed H. Alqahtani:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition. **Karar Mahmoud:** Writing – review & editing, Validation, Resources, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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